Brac University

CSE 421

Assignment - 02

Name - Farhan Hossain Sajid

ID - 20201129

sec - 24

1) (i) Network A has 1022 hosts needs /22 ($2^{10} - 2 = 1022$). PC's source IP is 172.16.8.1, which is the first usable host → Network A = 172.16.8.0/22 (range 172.16.8.0 - 172.16.11.255)

Network B is the next consecutive /22 subnet → Network B = 172.16.12.0/22

1) (ii) Maximum hosts in Network B

Network B is /22 → $2^{10} - 2$

= 1022 hosts maximum

1) (iii)

Subnet	Hosts needed	Prefix	Block size	Network address	Usable range	Broadcast
R1 LAN (1048 hosts)	1048	/22	1024 → need /21: $2^{11}-2=2046$	10.192.128.0/21	.129.135-.254	10.192.135.255
R2 LAN (250 hosts)	250	/24	$2^8-2=254$	10.192.136.0/24	.136.1. 136.254	10.192.136.255
SW2 LAN (62 hosts)	62	/26	$2^6-2=62$	10.192.137.0/26	.137.1. 137.62	10.192.137.63

2) (i) MTU value (X)

Fragment offset difference = 75. Fragment offset is measured in 8 byte units, so data per fragment = $75 \times 8 = 600$ bytes

MTU = data payload + IP header = 600 + 20

$X = 620$ bytes

2) (ii) Fragment offset of 4th fragment

offset increases by 75 per fragment (0-indexed fragment number). 4th fragment (index 3) → offset = $3 \times 75 = 225$

Fragment	1st	2nd	3rd	4th
offset	0	75	150	225

2) (iii) Data size of last fragment

total data = $4680 - 20 = 4660$ bytes

Full fragments: $4660 \div 600 = 7$ full fragments

$(7 \times 600 = 4200$ bytes$)$ + remainder =

$4660 - 4200 = 460$ bytes (last fragment data size)

Total fragments = 8

2) (iv) If the DF (Don't fragment) bit is set, the router cannot fragment the packet. If the packet exceeds the link MTU, the router drops it and sends an ICMP Type 3, Code 4 ("Fragmentation Needed and DF Set") message back to the source. The source would then use Path MTU Discovery (PMTUD) to reduce its segment size and retransmit smaller packets that fit within the MTU without fragmentation.

3)(i) Saba has no IP address yet. The first DHCP message is a DHCP Discover. The source IP address is 0.0.0.0 (all zeros, indicates the sender has no IP address assigned yet). The destination is 255.255.255.255 (broadcast).

3) (ii) Although both DHCP Discover messages are broadcast, each contains a Unique client Hardware Address field in the DHCP packet, which carries the client's MAC address. Saba's MAC = "SABA" and Asper's MAC = "ASPER". DHCP also uses a Transaction ID (xid), a random 32 bit number chosen by each client to match requests to responses. DHCP-1 uses these two field to distinguish and track each client's request independently.

3)(iii) Eve's lease from DHCP-2 has expired (renewal failed after 5 hours). She now has no valid IP address and reverts to 0.0.0.0. She sends a new DHCP Discover broadcast (src 0.0.0.0, dst 255.255.255.255). This broadcast crosses the router via DHCP relay (ip helper-address) configured on the router interface, which forwards the broadcast as a unicast to both DHCP servers. DHCP-1 receives the relayed Discover, offers an available IP from its pool and Eve completes the DORA process with DHCP-1 to obtain a new lease.

3) (iv) 1. ARP Request (broadcast): Asper sends "who has [Router R's IP]? Tell ASPER" - source MAC: ASPER, destination MAC: FF:FF:FF:FF:FF:FF

2. ARP Reply (unicast): Router R responds "Router R's IP is at [Router R's MAC]" - source MAC, Router R, destination MAC: ASPER.

After this, Asper encapsulates the IP packet in an Ethernet frame with destination MAC = Router R's MAC and forwards it.

4) (i) R2's LAN is 10.10.11.0/24. From R4, the path to R2 goes via R4's s1 (10.16.10.0/30) → R3's s2 (10.16.10.0/30)

From the table, R3's s2 IP with .2 = 10.16.10.2 and R4's s1 connects to R3 s2. The next-hop toward R2 from R4 is R3's s2 interface = 10.16.10.2, but for a non-recursive route we specify the exit interface (s1) and next-hop; R3's s2 interface = 10.16.10.2, but for a non-recursive route we specify the exit interface (s1) and next-hop;

R4(config)# ip route 10.10.11.0 255.255.255.0 s1 10.16.10.2

4) (ii) The current default route via s0 uses the Internet gateway. A floating static route has a higher administrative distance than the primary route (default AD for static = 1), so we assign AD > 1 (e.g 5 or higher)

R4(config)# ip route 0.0.0.0
0.0.0.0
s0 10.18.10.2 5

5) (i)

Compressed	Expanded (full 128 bit)
FE80::1A2B	FE80:0000:0000:0000:0000:0000:0000:1A2B
2001:DB8:1234::10	2001:0DB8:1234:0000:0000:0000:0000:0010
::25	0000:0000:0000:0000:0000:0000:0000:0025

5) (ii) 224.x.x.x is an IPv4 multicast address (Class D, 224.0.0.0/4). Unlike a broadcast, multicast packets are only delivered to hosts that have joined the multicast group 224.10.10.5 will receive the packet. Switches with IGMP snooping enabled will also forward the multicast only to ports where interested receivers are connected, rather than flooding to all ports.

6) (i) Asif does not know Eve's MAC address. He first sends an ARP Request. The Ethernet frame is:

- source MAC: ASIF | Destination MAC:

  FF:FF:FF:FF:FF:FF

  (broadcast)

Switch processing (all switches have empty MAC tables intially):

S2: Receives frame on Gi0/0 (from Asif). Learns ASIF → Gi0/0. Floods out Gi0/1 (toward S1/Eve) and Gi0/2 (toward S4).

S1: Receives on Gi1/1, Learns ASIF -> Gi1/1. Floods out Gi1/0 (Saba), Gi1/2 (S3/Fa0/2). Eve is on S2's Gi0/1 side - actually Eve connects directly to S2 Gi0/1, so S1 forwards to its other ports (Saba's port and toward S3)

S3: Receives on Fa0/2. Learns ASIF -> Fa0/2. Floods out Fa0/1 (toward Router R), Fa0/3 (Aspen)

S4: Receives on Gi0/2. Learns ASIF → Gi 0/2. Floods out Fa 1/0 (toward S4's other ports) and Fa 1/1 (DHCP_2).

Since all ports are flooded, Eve receives the ARP Request (she is connected to S2 Gi 0/1)

6) (ii) Eve sends an ARP reply (unicast, SRC MAC = EVE, dst MAC = ASIF).

Switches learn EVE's MAC on the incoming port and forward toward Asif:

Switch	MAC	Port
S2	ASIF	Gi 0/0
	EVE	Gi 0/1
S1	ASIF	Gi 1/1
	EVE	Gi 1/1 (reply came back same way)
S3	ASIF	Fa 0/2
	EVE	Fa 0/2
S4	ASIF	Gi 0/2
	EVE	Gi 0/2

7) (i) Beyond counting hops, TTL prevents routing loops from consuming bandwidth indefinitely. If a packet enters a routing loop (e.g caused by misconfiguration or convergence delay), each router decrements TTL. When TTL reaches 0, the router discards the packet and sends an ICMP Time Exceeded message back to the source, ensuring the packet is eventually destroyed

Without TTL: Looping packets would circulate forever, multiplying at every loop iteration. Routers would be overwhelmed processing and forwarding these "immortal" packets. Available bandwidth and CPU would be exhausted, bringing entire networks to a halt, a form of denial of service caused purely by misconfiguration.

7) (ii) This is a Smurf Attack, the attacker sends a single ICMP Echo Request with the source IP spoofed to the victim's IP and the destination set to the network's broadcast address (e.g 192.168.1.255). All 200 active devices on that broadcast domain receive the Echo Request and each replies with an ICMP Echo Reply, but they all send it to the spoofed source (the victim). The victim is thus flooded with 200 replies from a single crafted packet.

8) (i) NO, count-to-infinity will not occur in this scenario. count-to-infinity is specific to Distance Vector (DV) routing. While R1 & R3 use DV, R2 & R4 use Link state (LS).

(ii) When R1-R3 link fails, the network is partitioned into two separate segments with no remaining path between them.

9) (i) The PAT pool has address 220.20.20.17/30. Both Host A & Host D use the same public IP but different assigned port numbers

Host	inside Local (src IP:port)	inside Global (after PAT)
Host A	192.168.2.105:52157	220.20.20.17:<port_A> (e.g 1024)
Host D	192.168.2.106:52157	220.20.20.17:<port_D> (e.g 1025)

9) (ii) No, it is not possible with the current PAT setup. PAT is designed for outbound connections only. NAT/PAT entries are created dynamically when an inside host initiates a connection. Host C only knows the public IP 220.20.20.17 not which internal host or port it should target. No static NAT entry exits for Host A, so inbound packets from Host C would be dropped at the router (no matching translation entry).

ip nat inside source static tcp 192.168.2.105 <port> 220.20.20.17 <ext_port>